CHEMISTRY

Time: 3 Hrs. **Max. Marks: 70**

General Instructions:

*The question paper is divided into **four** sections.*

(1) ***Section A:*** *Q. No. **1** contains **Ten** multiple choice type of questions carrying **One** mark each. Only the first attempt will be considered for evaluation.*

*Q. No. **2** contains **Eight** very short answer type of questions carrying **One** mark each.*

(2) ***Section B:*** *Q. No. **3** to Q. No. **14** are **Twelve** short answer type questions carrying **Two** marks each. (Attempt **any Eight**)*

(3) ***Section C:*** *Q. No. **15** to Q. No. **26** are **Twelve** short answer type questions carrying **Three** marks each. (Attempt **any Eight**)*

(4) ***Section D:*** *Q. No. **27** to Q. No. **31** are **Five** long answer type of questions carrying **Four** marks each. (Attempt **any Three**)*

(5) *Use of log table is allowed. Use of calculator is not allowed.*

(6) *Figures to the right indicate full marks.*

(7) *Physical constants:*

i. Avogadro's Number = $N_A = 6.022 \times 10^{23}$

ii. Gas constant R = 0.08206 L – atm. K^{-1}. mol^{-1} / 8.314 JK^{-1}. mol^{-1}

iii. 1 Faraday = (F) = 96500 Coulombs

SECTION – A

Q.1. Select and write the correct answer for the following multiple choice type of questions: **[10]**

i. Which of the following is NOT an elastomer?

(A) vulcanized rubber (B) buna-S

(C) neoprene (D) polythene

ii. Anisole reacts with HI at 398K to form –

(A) cyclohexanol (B) benzenol

(C) iodobenzene (D) benzene

iii. Unit of Henry's constant is _____.

(A) mol L^{-1} bar^{-1} (B) mol L bar

(C) mol L^{-1} bar (D) mol^{-1} L^{-1} bar^{-1}

iv. The shape of bromine penta-fluoride is_____.

(A) Linear (B) Bent T Shape

(C) Tetrahedral (D) Square pyramidal

v. The relation between edge length and radius for body centered cubic structure is –

(A) $r = \frac{a}{2}$ (B) $r = \sqrt{3}\frac{a}{4}$

(C) $r = \sqrt{2}\frac{a}{4}$ (D) $r = \sqrt{3}\frac{a}{2}$

vi. Atomic number of post actinoid element range from –

(A) 58 – 72 (B) 78 – 92 (C) 93 – 103 (D) 104 – 118

vii. The pH of buffer solution containing 0.05 mol NaF per litre and 0.01 mol HF per litre (pKa = 3.1427) is _____.

(A) 3.8417 (B) 3.1427

(C) 0.3841 (D) 0.0384

viii. Preparation of ethanamine from acetonitrile represents –

(A) Gabriel phthalimide synthesis (B) Mendius reduction

(C) Hofmann hypobromite degradation (D) Sandmeyer reaction

ix. The source of dichlorocarbene is _____.

(A) methyl chloride (B) methylene chloride

(C) chloroform (D) carbon tetrachloride

x. The regent used to convert acetyl chloride to acetaldehyde is –

(A) $SnCl_2 + HCl$ (B) $Zn{\cdot}Hg$ + Conc.HCl

(C) $NH_2 - NH_2 + KOH/$ $\begin{matrix} CH_2 - OH \\ | \\ CH_2 - OH \end{matrix}$ (D) $H_2/Pd–BaSO_4$

Q.2. Answer the following questions: **[8]**

i. Write the relation between conductivity and molar conductivity.

ii. Write the oxidation state of Nickel in $[Ni(Cl)_4]^{2\ominus}$ complex ion

iii. Write the name of nanostructured material used in tyres of car to increase their life.

iv. Write the glycosidic linkage present in maltose.

v. Draw the structure of perchloric acid.

vi. Write the trade name of most common thermoplastic polyester.

vii. Calculate change over temperature for a certain reaction, whose $\Delta H°$ is – 60 kJ and $\Delta S°$ is – 150 JK^{-1}.

viii. Write the unit of rate constant of zero order reaction.

SECTION – B

Attempt any EIGHT of the following questions: **[16]**

Q.3. Write reactions involved during preparation of Nylon 6, 6 fibre.

Q.4. Derive Ostwald's dilution law equation for a weak acid.

Q.5. Draw the structure of two forms of alanine having pH $\leq$ 2 and pH $\geq$ 10.

Q.6. Define the following terms:

i. Intensive property

ii. Standard enthalpy of combustion

Q.7. What is the action of following reagents on ethyl amine?
(Indicate by reaction)

i. Nitrous acid

ii. Chloroform and KOH

Q.8. i. State Kohlrausch law of independent migration of ions.

ii. Write Nernst equation to calculate electrode potential.

Q.9. i. Write general electronic configuration of 3d series elements.

ii. Write any two points which explain about the colour of a transition metal ion.

Q.10. Explain aldol condensation reaction with respect to ethanal.

Q.11. Write a note on Kolbe reaction

Q.12. i. Write the relation between standard cell potential and equilibrium constant.

ii. Write the representation of sodium hexacyanoferrate (III) complex.

Q.13. In first order reaction 60% of reactant decompose in 40 min. Calculate the rate constant of the reaction.

Q.14. Explain optical activity of 2-chlorobutane.

SECTION – C

Attempt any EIGHT of the following questions: **[24]**

Q.15. i. Derive a relationship between molar mass, density of the substance and unit cell edge length.

ii. Define – paramagnetic solids.

Q.16. i. Mention the names of four principle of green chemistry.

ii. Write advantages of nanoparticles and nanotechnology in the field of energy sector.

Q.17. i. State second law of thermodynamics.

ii. Define – Nucleotides.

iii. Write chemical reaction for conversion of benzene diazonium chloride into benzene.

Q.18. The solubility product of AgBr is 5.2×10^{-13}. Calculate its solubility in mol dm^{-3} and g dm^{-3}. (Molar mass of AgBr = 187.8 g mol^{-1})

Q.19. Convert the following:

i. Ethanol to ethoxyethane.

ii. Phenol to p-benzoquinone

iii. Salicylic acid to aspirin

Q.20. i. State Raoult's law.

ii. Derive the relationship to determine molar mass of non volatile solute by depression of freezing point.

Q.21. i. Explain the formation of $[CoF_6]^{3\ominus}$ complex ion with respect to

a. Oxidation State of Cobalt

b. Type of hybridisation

c. Magnetic property

d. Geometrical shape

ii. Write the name of ligand used to estimate hardness of water.

Q.22. i. Write cell reactions of hydrogen-oxygen fuel cell at anode and cathode.

ii. Write an application of electrochemical series.

Q.23. i. Explain anomalous behaviour of oxygen with respect to

a. Atomicity

b. Magnetic property

c. Oxidation state

d. Nature of hydride

ii. Write any two uses of helium.

Q.24. What is the energy of activation of a reaction whose rate constant doubles when the temperature changes from 303 K to 313 K?

Q.25. i. Write chemical reactions taking place during extraction of iron in zone of reduction of the blast furnace.

ii. Write observed electronic configuration of gadolinium. (z = 64)

Q. 26 **Convert the following:**

i. Acetic acid to ethyl alcohol

ii. Acetic acid to acetic anhydride

iii. Acetyl chloride to acetamide

SECTION – D

Attempt any THREE of the following questions: **[12]**

Q.27. i. Define:
   a. Co-ordination numbers
   b. Ambidentate ligand
   ii. Write a note on Clemmensen reduction of propanone.

Q.28. i. Derive relationship between ΔH and ΔU for a gaseous chemical reaction
   ii. Give reason – Ozone is a good bleaching agent.
   iii. Write any two uses of SO_2.

Q.29. i. Define – Polymorphism.
   ii. State and explain Hess's law of constant summation. Write its one application.

Q.30. i. 0.01 m aqueous formic acid solution freezes at – 0.021 °C. Calculate its degree of dissociation. ($K_f = 1.86$ K. kg. mol^{-1})
   ii. Write properties of interstitial compounds.

Q.31. i. What is the action of following regents on Bromoethane?
   a. KNO_2
   b. 'Mg' metal
   c. Silver acetate
   ii. Draw structure – DDT insecticide.